# Mamdani Fuzzy Inference System

## 1. Problem Description

An autonomous delivery robot must decide its driving speed in uncertain conditions. Sensor readings for visibility (0–100%), obstacle distance (0–10 m), and battery level (0–100%) are imprecise, so rigid if/else rules are not suitable. A Mamdani fuzzy inference system is used instead, producing a crisp recommended speed (0–10 km/h) via centroid defuzzification.

## 2. Input and Output Variables

| Variable | Range | Unit | Fuzzy sets |
|---|---|---|---|
| Visibility | 0 – 100 | % | low, medium, high |
| Obstacle distance | 0 – 10 | m | close, medium, far |
| Battery level | 0 – 100 | % | low, medium, high |
| Recommended speed (output) | 0 – 10 | km/h | very slow, slow, moderate, fast |

## 3. Rule Base

The system has 9 rules (8 original + 1 added in Task 1):

1. IF visibility is low OR obstacle distance is close THEN speed is very slow
2. IF visibility is medium AND obstacle distance is close THEN speed is very slow
3. IF visibility is medium AND obstacle distance is medium THEN speed is slow
4. IF visibility is high AND obstacle distance is medium THEN speed is moderate
5. IF visibility is high AND obstacle distance is far AND battery is high THEN speed is fast
6. IF battery is low AND obstacle distance is far THEN speed is moderate
7. IF battery is low AND obstacle distance is close THEN speed is very slow
8. IF visibility is medium AND battery is medium AND obstacle distance is far THEN speed is moderate
9. IF visibility is high AND battery is low AND obstacle distance is far THEN speed is moderate [Task 1]

Rule 1 uses OR because either condition alone is enough to require slowing down.

Rule 5 requires all three conditions to be favourable before allowing fast speed.

Rule 6 shows that low battery alone is a reason to limit speed even in safe conditions.

## 4. Mamdani Inference Mechanism

- Fuzzification: each crisp input is mapped to membership degrees across all fuzzy sets using triangular or trapezoidal functions.
- Rule activation: antecedent strength computed using min() for AND and max() for OR.
- Implication: consequent set is clipped at the rule strength using min(strength, set).
- Aggregation: all clipped sets are combined with element-wise max().
- Defuzzification: crisp output computed as centroid = sum(x * mu(x)) / sum(mu(x)).

## 5. Experiments

| Visibility (%) | Distance (m) | Battery (%) | Mamdani (km/h) | Crisp (km/h) |
|---|---|---|---|---|
| 20 | 1.0 | 90 | ~1.3 | 2.0 |
| 90 | 9.0 | 90 | ~7.8 | 8.0 |
| 70 | 4.0 | 30 | ~3.5 | 4.0 |
| 45 | 7.5 | 55 | ~4.2 | 5.0 |
| 80 | 2.0 | 80 | ~1.6 | 2.0 |
| 39 | 5.0 | 80 | ~3.8 | 2.0 |
| 40 | 5.0 | 80 | ~3.8 | 5.0 |

The last two rows highlight the key difference: the crisp system jumps from 2.0 to 5.0 between visibility 39 and 40, while Mamdani stays nearly the same.

## 6. Critical Analysis

The system handles mid-range inputs well, where multiple rules fire and balance each other. It struggles near membership function boundaries , as shown in Task 2, widening the "close" set from (0,0,1.5,3.5) to (0,0,2.5,5.0) noticeably lowered the speed for distance = 4 m without any real change in the physical situation. The rule base is also small for a three-input system, so some input combinations are not well covered.

## 7. Possible Improvements

- Calibrate membership function boundaries using real sensor data instead of choosing them manually.
- Extend the rule base to better cover mixed input combinations.
- Add inputs like terrain type or speed of the approaching obstacle.
- Validate against expert-defined reference speeds in a simulated environment.

## Discussion Questions

1. Why is a fuzzy rule-based system useful in uncertain environments?

Fuzzy systems handle imprecise inputs gracefully. Hard thresholds cause abrupt jumps; fuzzy membership allows gradual, safer transitions.

2. What is the role of membership functions?

Membership functions map a crisp value to a degree of belonging to a linguistic category, allowing partial membership in multiple sets at once.

3. What does it mean for a rule to be partially activated?

A rule is partially activated when its antecedent strength is between 0 and 1, meaning it contributes to the output proportionally rather than fully or not at all.

4. Why does Mamdani inference need an aggregation step?

Aggregation is needed because multiple rules can fire simultaneously and target the same output variable. Max combines their clipped sets into one.

5. Why is defuzzification necessary?

Defuzzification converts the aggregated fuzzy set into a single number that can actually be used to set the motor speed.

6. What are the limitations of this system?

Limitations: small rule base, manually chosen membership functions, no temporal awareness, sensitivity to boundary placement.

7. How could this system be validated in a real robotic application?

Validation: run the robot in known scenarios, record recommended speeds, and compare against expert reference values or a ground-truth controller.